Nextcloud Server Automation

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Technology Stack

Ubuntu Server – Lightweight and stable base OS for server deployment

Nextcloud – Self-hosted file storage, collaboration, and productivity suite.

Snap – Used for simplified installation, updates, and maintenance of Nextcloud.

Tailscale VPN – Mesh VPN solution for secure remote access outside the local network.

Project Overview

Welcome to my project!

I‘m honored that you are visiting my portfolio website and taking the time to explore my work. In this project, I set up a self-hosted Nextcloud server on Ubuntu using the Snap package manager, which simplified the installation and configuration process. Initially, the server was accessible only within the local network. To enable secure remote access, I implemented HTTPS and configured Tailscale VPN. This setup ensures encrypted, private access to the server from outside the local network, enhancing both data accessibility and collaboration capabilities.

Implementation Steps

After booting the system from the bootable USB stick containing Ubuntu Server, I began the installation process. The first step was to select the system language and keyboard layout, as shown in Figure 1.

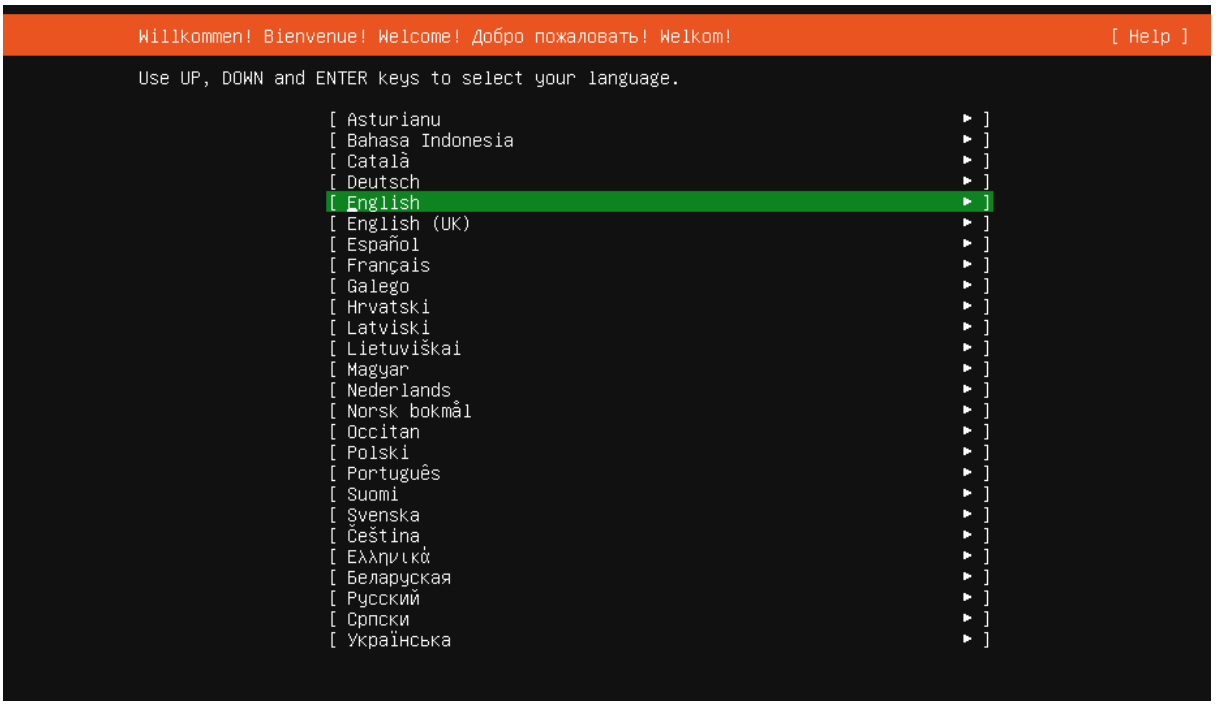

Figure 1: Language and keyboard layout selection

In the next step, I proceeded with the network configuration. I configured the server to operate within the subnet 192.168.178.0/24. The server was assigned the static IP address 192.168.178.50. As the gateway, I specified the IP address of my router: 192.168.178.1. For the name server, I used Google's public DNS server at 8.8.8.8. This DNS server is widely used due to its reliability, speed, and global availability, making it a solid choice for resolving domain names to IP addresses. See Figure 2.

Figure 2: Network configuration with static IP

For the partitioning step, I selected the option to use the entire disk. This simplifies the setup by automatically allocating all available space for the installation. In the next screen, a summary of the file systems to be installed on the disk was displayed, showing the partition layout and mount points. See Figure 3 and 4.

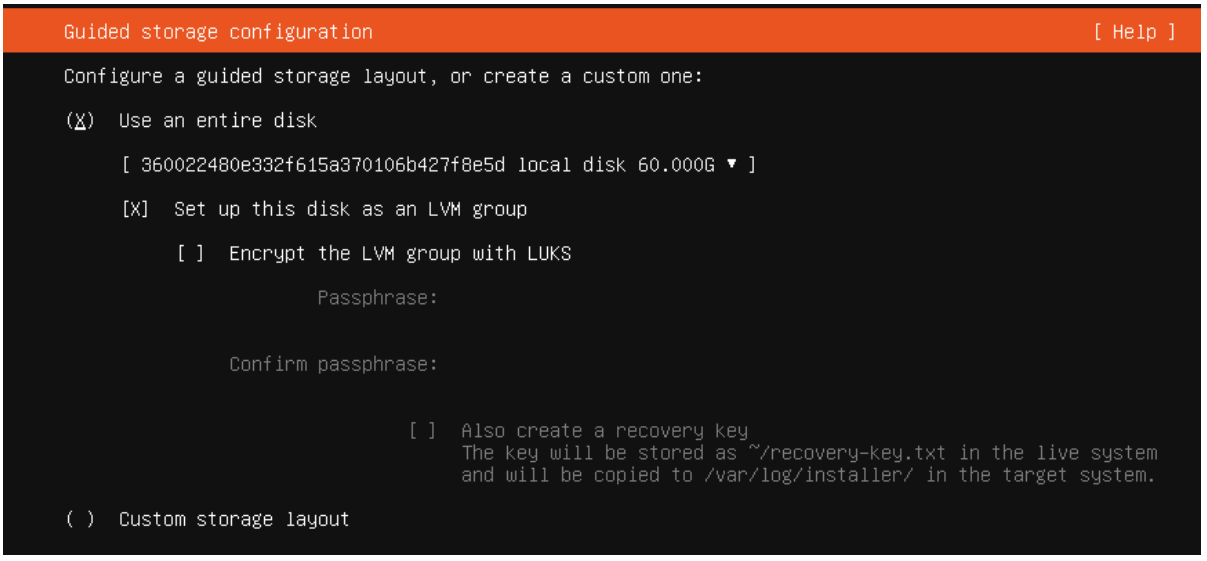

Figure 3: Disk partitioning - Use entire disk option

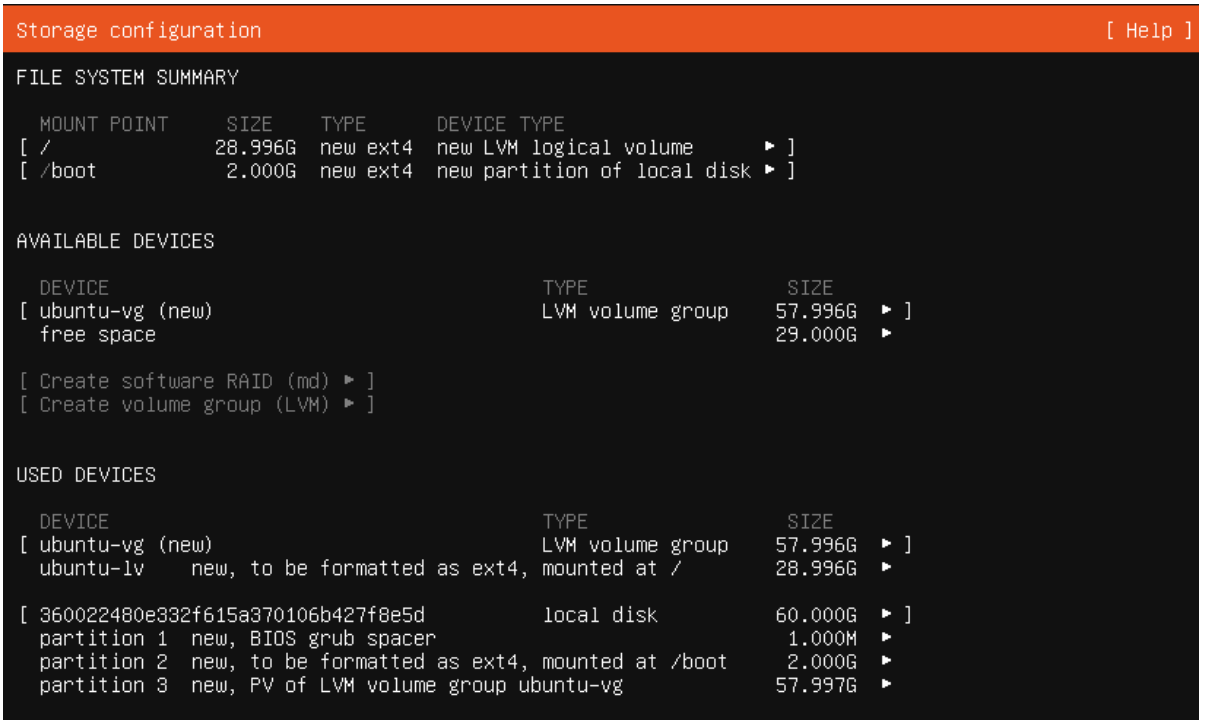

Figure 4: File system summary before installation

For the profile configuration, I set the username to my name, Mahmoud □, since I will likely be the only one administering the server. The hostname of the machine is

set to nextcloud. After that, I created a secure password for the user account. See Figure 5.

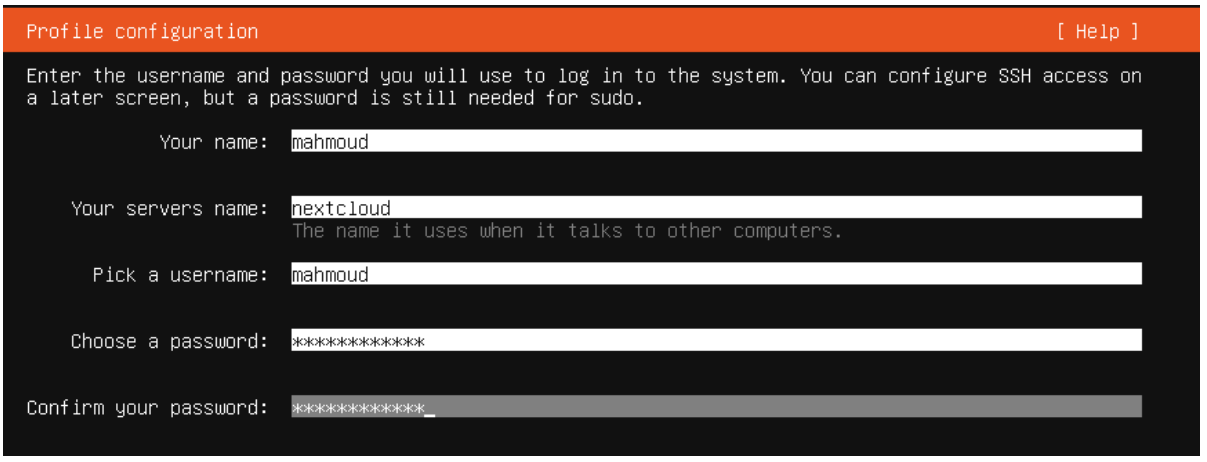

Figure 5: Hostname, username, and password setup

In the following step, I chose to install OpenSSH. This allows secure remote access to the server via SSH, which is essential for managing the server remotely in the future. See Figure 6.

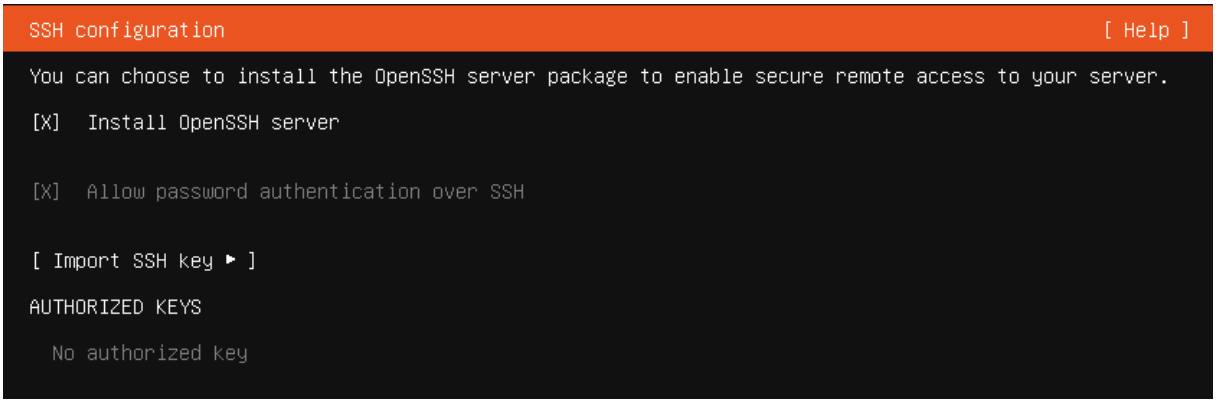

Figure 6: OpenSSH installation for remote server access

After completing the configuration steps, the installation process began. The system started copying files and setting up the selected components. See Figure 7.

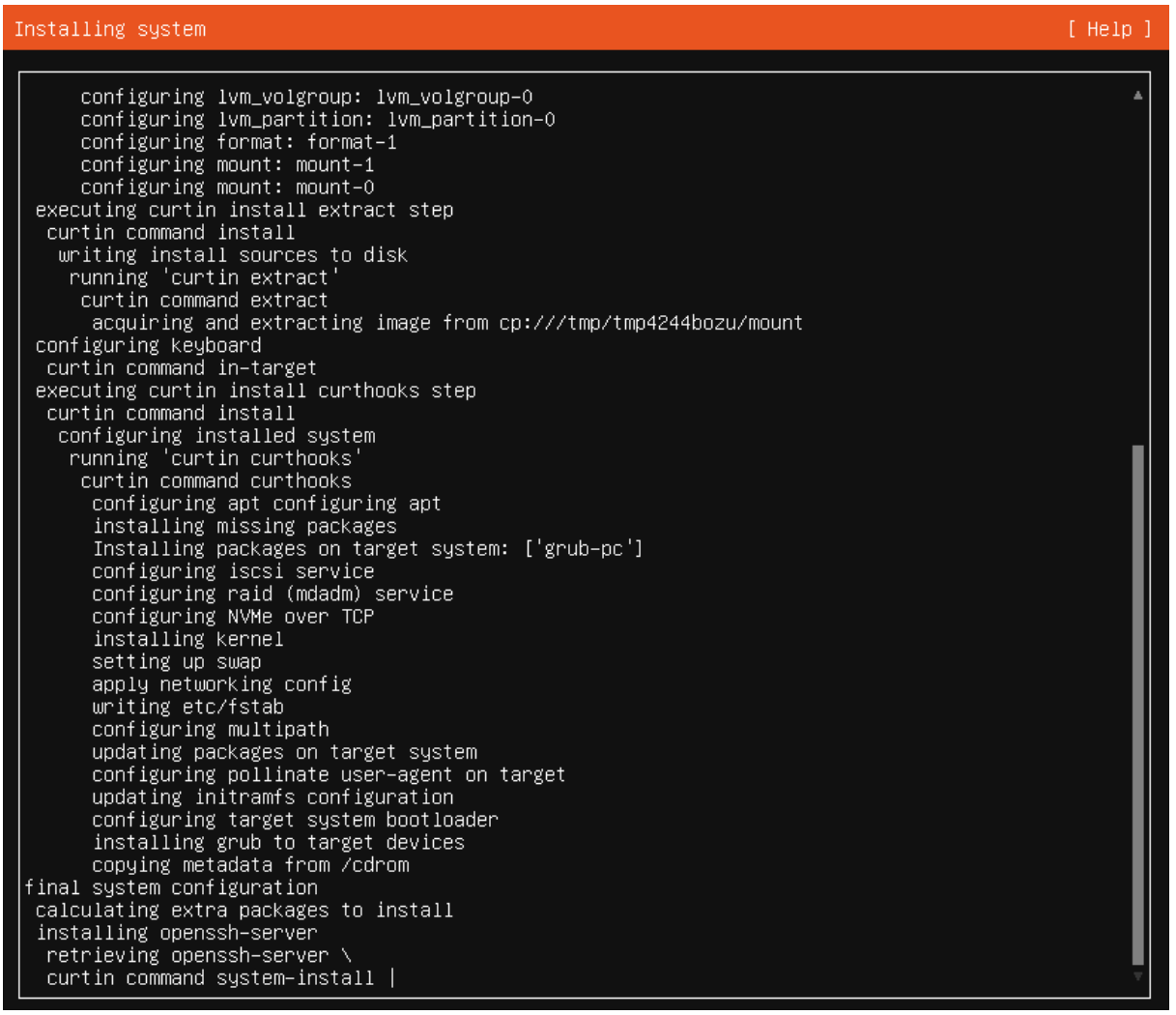

Figure 7: Ubuntu Server installation in progress

Once the installation was complete and the server has been rebooted, I switched to my computer and connected to the server using SSH. I ran the command: ssh 192.168.178.50. After entering the previously defined password, I successfully gained remote access to the server. After gaining remote access to the server via SSH, I updated the system to ensure all packages were up to date. I ran the following command: sudo apt update && sudo apt full-upgrade -y. This command updates the packages list and installs the last versions of all installed packages. In the screenshot, you can see the message “All packages are up to date”, because I had already performed this update step earlier during preparation. As a result, the system was already fully upgraded at the time of documentation. See Figure 8.

```
mahmoud@nextcloud:~$ sudo apt update && sudo apt full-upgrade -y
[sudo] password for mahmoud:
Hit:1 http://de.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://de.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://de.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
mahmoud@nextcloud:~$
```

Figure 8: System update

Next, I installed Nextcloud using Snap package manager on the Ubuntu server. To do this, I ran the following command in the terminal: sudo snap install nextcloud. See figure 9.

```
mahmoud@nextcloud:~$ sudo snap install nextcloud
2025-06-15T14:55:48Z INFO Waiting for automatic snapd restart...
Download snap "nextcloud" (48223) from channel "stable"
nextcloud 31.0.5snap1 from Nextcloud✓ installed
mahmoud@nextcloud:~$
```

Figure 9: Nextcloud installation with Snap

After the installation, I opened a web browser and entered the server's IP address, http://192.168.178.50, in the address bar. This directed me to the Nextcloud login page. I then created an account by using the username root since I am the administrator and set a password. See figure 10.

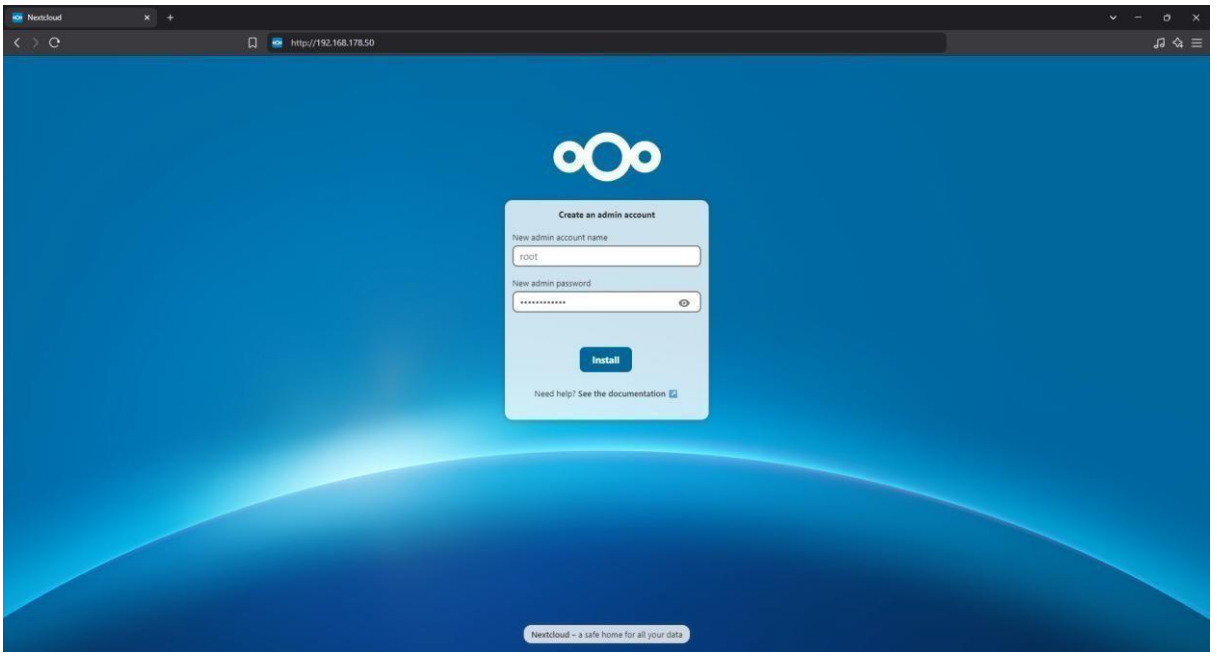

Figure 10: The login page of nextcloud

To make Nextcloud accessible not only within the local network but also from outside, I installed Tailscale on the Ubuntu server. I installed Tailscale using the following command: curl -fsSL https://tailscale.com/install.sh | sh. Then, I activated it with: sudo tailscale up. See figure 11 and 12. Before installing Tailscale, I enabled HTTPS in Nextcloud using the command: sudo nextcloud.enable-https lets-encrypt.

```
mahmoud@nextcloud:~$ curl -fsSL https://tailscale.com/install.sh | sh
Installing Tailscale for ubuntu noble, using method apt
+ sudo mkdir -p --mode=0755 /usr/share/keyrings
[sudo] password for mahmoud:
+ curl -fsSL https://pkgs.tailscale.com/stable/ubuntu/noble.noarmor.gpg
+ sudo tee /usr/share/keyrings/tailscale-archive-keyring.gpg
+ sudo chmod 0644 /usr/share/keyrings/tailscale-archive-keyring.gpg
+ curl -fsSL https://pkgs.tailscale.com/stable/ubuntu/noble.tailscale-keyring
.list
+ sudo tee /etc/apt/sources.list.d/tailscale.list
# Tailscale packages for ubuntu noble
```
Figure 11: The installation of Tailscale

```
mahmoud@nextcloud:~$ sudo tailscale up

To authenticate, visit:

        https://login.tailscale.com/a/1f19b1c401aba8
```
Figure 12: Tailscale activation

After running sudo tailscale up, a link was displayed in the terminal. I copied this link and opened it in a browser to log in or create an account if I didn't have one already. But I

already have an account, and I logged in and entered the admin console of Tailscale as shown in figure 13. This step connected the server to my Tailscale VPN network. Once connected, I was able to access Nextcloud via the server's Tailscale IP address from any device connected to the same Tailscale network.

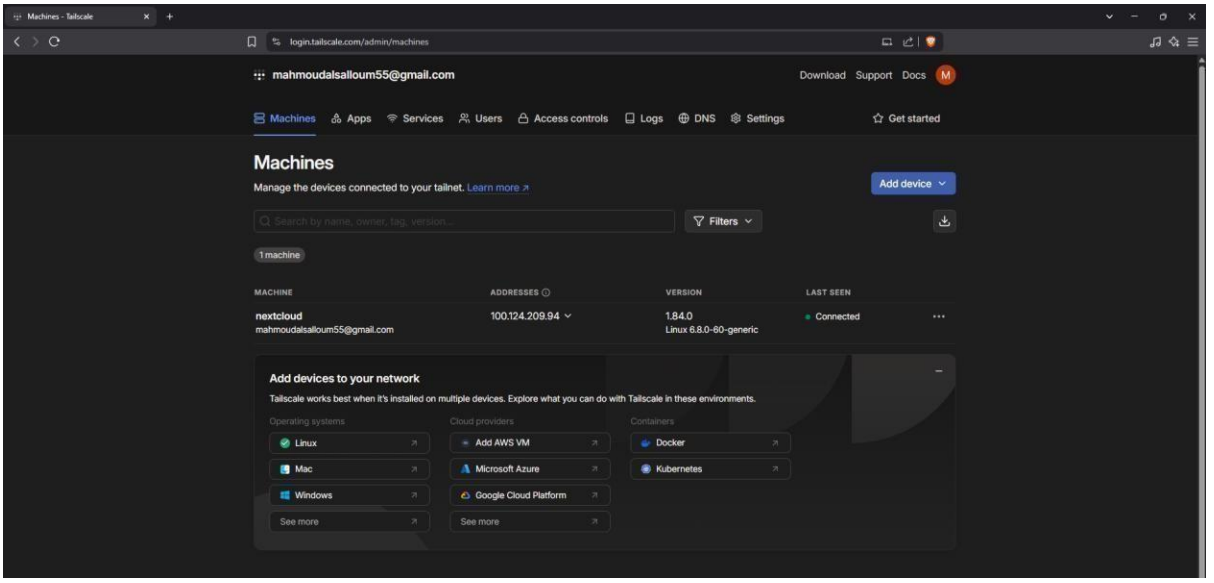

Figure 13: Tailscale admin console

I logged into Tailscale on my smartphone using my account and joined the Tailscale network. One connected, my smartphone was immediately visible in the Tailscale admin. See figure 14 and 15.

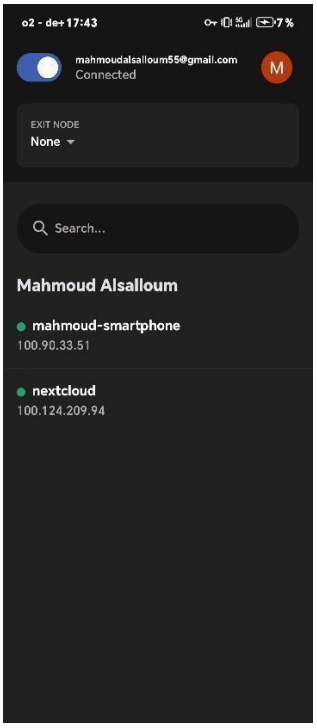

Figure 14: My smartphone joins Tailscale network

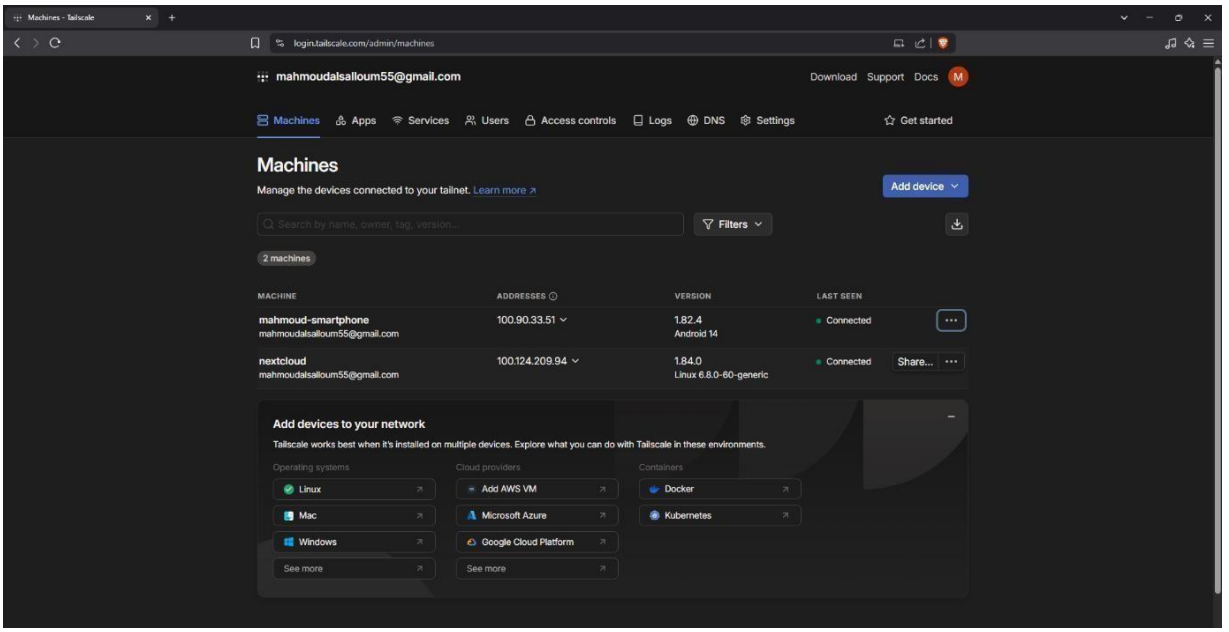

Figure 15: My smartphone is in Tailscale admin console

After installing the Nextcloud app on my smartphone, I tried to access the server using its Tailscale IP address. However, I encountered an error message saying “Access through untrusted domain.This occurred because Nextcloud, for security reasons, only allows connections from domains or IP addresses that are explicitly defined as trusted. Since the Tailscale IP was not yet added to the trusted domains list, the connection was blocked. See Figure 16.

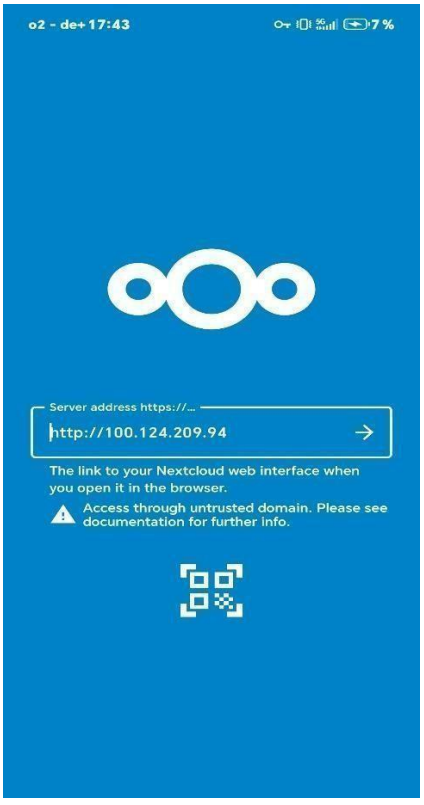

Figure 16: The error message of Nextcloud

To fix the “Access through untrusted domain” error, I edited Nextcloud’s configuration file to add the Tailscale IP address as a trusted domain. I opened the configuration file using the following command: sudo nano /var/snap/nextcloud/current/nextcloud/config/config.php

In the ‘trusted_domains’ section of the file, I added a new entry for the Tailscale IP address of my server, as shown below in Figure 17.

```
GNU nano 7.2            /var/snap/nextcloud/current/nextcloud/config/config.php *
      'path' => '/snap/nextcloud/current/htdocs/apps',
      'url' => '/apps',
      'writable' => false,
    ),
    1 =>
    array (
      'path' => '/var/snap/nextcloud/current/nextcloud/extra-apps',
      'url' => '/extra-apps',
      'writable' => true,
    ),
  ),
  'supportedDatabases' =>
  array (
    0 => 'mysql',
  ),
  'memcache.locking' => '\\OC\\Memcache\\Redis',
  'memcache.local' => '\\OC\\Memcache\\Redis',
  'redis' =>
  array (
    'host' => '/tmp/sockets/redis.sock',
    'port' => 0,
  ),
  'log_type' => 'file',
  'logfile' => '/var/snap/nextcloud/current/logs/nextcloud.log',
  'logfilemode' => 416,
  'maintenance_window_start' => 1,
  'instanceid' => 'oc0k6ia3ssmb',
  'passwordsalt' => 'r51E73cVi1/7NGZgilhKEzv/CpYkXx',
  'secret' => '78Bh/R2Kh1suoGkRrlFsHF1U21WxFQwxNzXkQV3U3PFrAEYY',
  'trusted_domains' =>
  array (
    0 => '192.168.178.50',
    1 => '100.124.209.94',
  ),
  'datadirectory' => '/var/snap/nextcloud/common/nextcloud/data',
  'dbtype' => 'mysql',
  'version' => '31.0.5.1',
  'overwrite.cli.url' => 'http://192.168.178.50',
  'dbname' => 'nextcloud',
  'dbhost' => 'localhost:/tmp/sockets/mysql.sock',
  'dbport' => '',
  'dbtableprefix' => 'oc_',
  'mysql.utf8mb4' => true,
  'dbuser' => 'nextcloud',
  'dbpassword' => 'emTG0K6QAEuRxEs0TrjJgButesq2bsG3LDqTE0soYXYoXxXhyU7vDPzBSl-g82xi',
  'installed' => true,
);

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

Figure 17: The solution to Nextcloud error

After saving the changes and closing the file, I restarted the Nextcloud web server with the following command:

sudo snap restart nextcloud.apache.

This resolved the issue, and I was then able to access Nextcloud from my smartphone using the Tailscale IP address without any errors. See Figure 18.

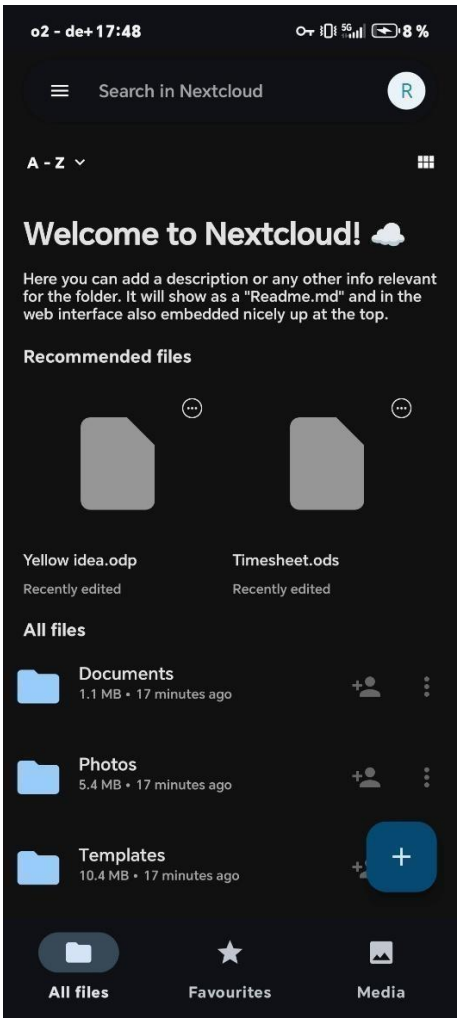

Figure 18: Nextcloud Access